

Algorithmic Trading

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HW 07: Python Pre-Trade Analysis

Assignment:

- Develop Python TCA Pre-Trade functions to calculate:
 - Market Impact (MI)
 - Timing Risk (TR)
 - Price Appreciation (PA)
 - POV Rate to Trade Time (PovToTime)
 - Trade Time to POV Rate (TimeToPOV)
 - Expected Cost (ExpCost)
 - Expected Execution Price (ExpPrice)
- All questions must be answered using your Python code.
- You must upload your Python code with your user-defined functions.
- See “TCA_Important Formulas.pdf” for all formulas.

MI Parameters

Use the following MI Parameters to answer all questions:

a1=883.58722; a2=0.35408; a3=0.755684; a4=0.826155; b1=0.963532

Q1) Pre-Trade Calculations:

Calculate MI, TR, PA, and Trade Time for the following order:

Side=Buy, Size=10%, Volatility=35%, AlphaBp=30bp, and POV=20%.

MI=51.53; TR=80.82; PA=6.0; POV= 20%, TradeTime= 0.40

Q2) Pre-Trade Calculations:

Calculate MI, TR, PA, and Trade Time for the following order:

Side=Sell, Shares = 125,000, ADV = 1,000,000, Volatility=40%, AlphaBp=10bp, and POV=10%.

MI=38.16; TR=154.92; PA=-5.62; POV= 10%, TradeTime= 1.125

Q3) Expected Cost and Expected Price

Calculate the Expected Cost and Expected Price, and all TCA cost components (MI, TR, PA, and POV) for the following order:

Side=Buy, Size=5%, Volatility=25%, AlphaBp=10bp, Price=75, and Trade Time = 1-day.

MI=12.27; TR=91.28; PA=5.0; POV= 4.76%, TradeTime= 1.0; ExpCost=17.27; ExpPrice=75.1295

Q4) Expected Cost and Expected Price

Calculate the Expected Cost and Expected Price and all TCA cost components (MI, TR, PA, and POV) for the following order:

Side=Sell, Shares=75,000, ADV=650,000, Volatility=30%, AlphaBp= -10bp, Price = 250 and Trade Time = 0.5 days.

MI=46.06; TR=77.45; PA=-2.5; POV= 18.75%, TradeTime= 0.5; ExpCost=48.5606;
ExpPrice=248.7859

Q5) Expected Cost and Expected Price

Calculate the Expected Cost and Expected Price and all TCA cost components (MI, TR, PA, and POV) for the following order:

Side=Buy, Shares=115,250, ADV=800,000, Volatility=50%, AlphaBp= -5bp, Price = 100 and Trade Time = 0.75 days.

MI=65.80; TR=158.11; PA=-1.875; POV= 16.11%, TradeTime= 0.75; ExpCost=63.93;
ExpPrice=100.6393

Q6) Expected Cost and Expected Price

Calculate the Expected Cost and Expected Price and all TCA cost components (MI, TR, PA, and POV) for the following order:

Side=Buy, Shares=100,000, ADV=2,000,000 Volatility=40%, AlphaBp=0bp, Price = 100 and POV=25%.

MI=52.49; TR=56.56; PA=0; POV= 25%, TradeTime= 0.15; ExpCost=52.49; ExpPrice=100.52.

Q7) Trade Time

Your portfolio manager gives you a buy order for 25% ADV and asks you to calculate the order using a POV=10% strategy. How long will it take you to complete the order under normal market conditions? E.g., What is the expected Trade Time for this order:

Trade Time = 2.25 Days

Q8) POV Rate

Your Portfolio manager gives you a sell order for 20% ADV and asks you to execute the order over T=0.75 trading days. What POV rate must to specify in the algorithm?

POV= 21.05%

Q9) Maximum Shares

What is the maximum number of shares of stock XYZ that you can buy in one day up to a market impact cost of MI=75bp? The characteristics of XYZ are: ADV=2,000,000 and Volatility=50%. There is no expected price trend in the stock. Shares = 396,099; MI=74.99995; Size=19.80%

Q10) Trade Time from AlphaBp

Stock XYZ opens at Open=\$65.00. The expected price trend over the day is AlphaBp =100bp. How long will it take for the price to increase to \$65.45? Time = 0.6924 days